



Principles of Finance

Chapter 10

End-of-Chapter Questions

Bonds and Bond Valuation

Multiple Choice Questions

1. A bond that has a coupon rate less than prevailing interest rates will _____.

- A. sell at par value
- B. sell at a discount
- C. sell at a premium
- D. be overpriced

Solution: B. A bond that has a coupon rate less than prevailing interest rates will sell at a discount (below par or face value).

Section 10.2 LO 2 Easy

2. Bond laddering is _____.

- A. a risky bond investment strategy that may yield tremendous returns
- B. a strategy in which bonds with several different maturity periods are added to a portfolio
- C. a strategy that involves replacing equity investments with bonds in a portfolio
- D. a bond strategy that sacrifices diversity for potential capital gains

Solution: B. Bond laddering is a strategy in which bonds with several different maturity periods are added to a portfolio in order to reduce risk and increase diversity of the asset mix.

Section 10.4 LO 4 Moderate

3. A call feature _____.

- A. is desirable to an investor
- B. may cause additional risk for the bond issuer
- C. may cause additional risk for an investor
- D. Both A and B

Solution: C. A bond with a call provision may present additional risk to an investor.

Section 10.4 LO 1 Moderate

4. When solving bond problems relating to a bond that pays interest on a quarterly basis, the _____ before being applied.
- A. quoted annual yield to maturity should be multiplied by 4
 - B. quoted number of years until maturity should be divided by 4
 - C. quoted annual coupon payments should be divided by 4
 - D. stated face value should be divided by 4

Solution: C. When solving bond problems, if an annual coupon payment is given, this should be divided by 4 to correspond to what the bondholder will receive in payment on a quarterly basis.

Section 10.1 LO 1 Moderate

5. A normal yield curve will _____.
- A. slope downward as it moves along its x-axis (term).
 - B. slope upward as it moves along its x-axis (yield).
 - C. fluctuate depending on the federal funds rate
 - D. slope upward as it moves along its x-axis (term).

Solution: D. A normal yield curve will slope upward as it moves further along its x-axis, which represents the term.

Section 10.3 LO 2 Moderate

6. Determining bond prices often involves using which two TVM (time value of money) equations?
- A. The future value of a lump sum and the present value of a lump sum
 - B. The present value of an annuity and the future value of a lump sum
 - C. The future value of an annuity and the present value of a lump sum
 - D. None of the above

Solution: D. None of the above. In order to determine the price of a bond, the present value of an annuity and the present value of a lump sum must both be calculated and then added together.

Section 10.2 LO 1 Easy

7. Which of the following is NOT an adjustment that must be made when interest is paid semiannually instead of annually?
- A. Dividing the annual coupon payment by 2
 - B. Dividing the annual interest rate by 2
 - C. Dividing the total number of years by 2
 - D. Dividing the annual yield to maturity by 2

Solution: C. The total number of years must be multiplied, not divided, by 2. This converts years into semiannual periods.

Section 10.1 LO 1 Moderate

8. The duration of a bond is _____.

- A. a measurement of the bond's overall risk
- B. synonymous with the bond's term
- C. a measurement of how long an investor holds the bond
- D. a measurement of the bond's sensitivity to interest rate changes

Solution: D. The duration of a bond is a measurement of the bond's sensitivity to interest rate changes.

Section 10.4 LO 1 Easy

9. Which of the following is NOT considered a factor that influences a bondholder's required rate of return?

- A. Financial risk
- B. Other investments by the bondholder
- C. Risk premium
- D. Business risk

Solution: B. Other investments by the bondholder will not typically influence the required rate of return.

Section 10.1 LO 1 Easy

10. How might an investment in a bond fund be affected by a decline in interest rates?

- A. The fund investment would not be affected.
- B. The fund investment would likely decrease in value.
- C. The fund investment would likely increase in value.
- D. Coupon payments from bonds in the fund would decline.

Solution: C. When interest rates drop, bond values increase. This means a bond fund (a mutual fund heavily invested in bonds) would experience a general increase in value if interest rates declined.

Section 10.1 LO 3 Easy

11. Interest rates and bond prices _____.

- A. are unrelated
- B. have an inverse relationship
- C. have a direct relationship
- D. are both economic factors set by central banks

Solution: B. Interest rates and bond prices have an inverse relationship.

Section 10.1 LO 3 Easy

12. The coupon rate of a bond is typically _____.

- A. fixed at the time of bond issuance
- B. subject to change based on the federal funds rate
- C. zero in the case of zero-coupon bonds
- D. Both A and C

Solution: D. Both A and C are correct. Coupon rates for standard bonds are fixed at the time a company or government issues them and will not change. Zero-coupon bonds are bonds with no periodic coupon or interest payments.

Section 10.1 LO 1 Easy

13. An inverted yield curve is an indication that _____.

- A. long-term yields and interest rates are higher than short-term rates
- B. the economy is in the process of a significant recovery
- C. short-term yields and interest rates are higher than long-term rates
- D. the yields to maturity on all bonds are less than market interest rates

Solution: C. An inverted yield curve indicates that short-term rates are higher than long-term rates.

Section 10.3 LO 3 Moderate

14. A zero-coupon bond is a bond that _____.

- A. has no value
- B. has no periodic coupon payments
- C. has been rated below investment grade
- D. Both A and C

Solution: B. A zero-coupon bond makes no periodic interest (coupon) payments.

Section 10.1 LO 2 Easy

Review Questions

1. Describe reinvestment risk for a bondholder.

Solution: Reinvestment risk is the risk that a bond investor will be unable to reinvest their current cash holdings at a similar rate of return to what they enjoyed with a previously held bond or investment. This can occur when prevailing interest rates drop and companies move to save money by calling their bonds in order to issue new bonds with lower payouts to investors.

Section 10.4 LO 1 Moderate

2. What is a junk bond?

Solution: A junk bond is a form of debt that has been given a low credit rating or is below investment grade. Junk bonds are considered riskier than other bonds because the possibility that the issuer will default or experience a credit event can be relatively high.

Section 10.1 LO 3 Moderate

3. Briefly describe interest income within the context of bond investments.

Solution: Within the context of bond investments, interest income refers to the coupon payments that bondholders receive in exchange for lending funds to the bond issuer. The interest rate that determines the coupon payment is referred to as the coupon rate.

Section 10.1 LO 1 Moderate

4. What is an inverted yield curve, and what is its significance?

Solution: An inverted yield curve occurs when short-term yields are higher than long-term yields, causing the yield curve to slope downward. An inverted yield curve indicates that investors believe that interest rates will fall in the near future, signaling a possible economic recession.

Section 10.3 LO 3 Moderate

5. Briefly describe the two types of cash flow that a bondholder will receive from the issuer.

Solution: A bondholder will receive both payment of the bond's par value at maturity (often referred to as payment of the face value of the bond at term end) and periodic coupon payments (also called interest income) from the bond issuer. Both the par value of the bond and the coupon payments are contractually determined and clearly indicated in the bond issue documentation received by the bondholder upon purchase.

Section 10.1 LO 1 Moderate

Problems

1. A \$1,000 Expo Corp. bond has a coupon rate of 5%, pays interest semiannually, and matures in six years. If the yield to maturity is 7%, what is the bond's value today?

Solution: $FV = \$1,000$

$$PMT = \frac{\$1,000 \times 5\%}{2} = \$25$$

$$N = 6 \times 2 = 12$$

$$I/Y = \frac{7\%}{2} = 3.5\%$$

$$PV = \$903.37$$

Section 10.2 LO 1 Moderate

2. A \$1,000 Omega Corp. bond has an 8% coupon rate that is paid semiannually. The bond matures in three years. If the current price of the bond is \$1,125, what is the yield to maturity?

Solution: $FV = \$1,000$

$$PMT = \frac{\$1,000 \times 8\%}{2} = \$40$$

$$N = 3 \times 2 = 6$$

$$PV = \$1,125$$

$$I/Y = 1.7846\% \text{ (per semiannual period)}$$

$$\text{YTM (must be annualized)} = 1.7846\% \times 2 = 3.57\%$$

Section 10.2 LO 1 Moderate

3. You are considering buying a bond that is currently priced at \$830, has a face value of \$1,000, and matures in seven years. If interest is paid semiannually and the bond has a yield to maturity of 6%, what is the bond's annual coupon rate?

$$\text{Solution: PV} = \$830$$

$$\text{FV} = \$1,000$$

$$N = 7 \times 2 = 14$$

$$I/Y = \frac{6\%}{2} = 3\%$$

$$\text{PMT (semiannual)} = \$14.9505$$

$$\text{PMT (annual)} = \$14.9505 \times 2 = \$29.90 \text{ per year}$$

$$\text{Annual coupon rate} = \frac{\$29.90}{\$1,000} = 2.99\%$$

Section 10.2 LO 1 Difficult

4. A \$1,000 Noah Corp. bond has a coupon rate of 5% with semiannual payments, matures in 10 years, and has a yield to maturity of 6.5%. What is the bond's current price?

$$\text{Solution: PMT} = \frac{\$1,000 \times 5\%}{2} = \$25$$

$$\text{FV} = \$1,000$$

$$N = 10 \times 2 = 20$$

$$I/Y = \frac{6.5\%}{2} = 3.25\%$$

$$\text{PV (current price)} = \$890.95$$

Section 10.2 LO 1 Moderate

5. Chronowerx Inc. has issued a bond that has a face value of \$1,000, a 3% coupon rate (with semiannual interest), and a maturity date four years from now. If the bond's current price is \$895, what is its yield to maturity?

$$\text{Solution: PMT} = \frac{\$1,000 \times 3\%}{2} = \$15$$

$$\text{FV} = \$1,000$$

$$N = 4 \times 2 = 8$$

$$\text{PV} = -\$895$$

$$I/Y = 2.9955\%$$

$$\text{Annualized YTM} = 2.9955\% \times 2 = 5.99\%$$

Section 10.2 LO 1 Difficult

6. You are considering adding a \$1,000, 25-year bond to your portfolio. It has a coupon rate of 8%, which is paid annually, and your required return is 10%. What is the current price of the investment?

Solution: $PMT = \$1,000 \times 8\% = \80

$FV = \$1,000$

$N = 25$

$I/Y = 10\%$

$PV \text{ (current price)} = \818.46

Section 10.2 LO 1 Moderate

7. A Cameron Corp. bond has a \$1,000 par value, a 5 percent coupon rate paid semiannually, and nine years until maturity. If similar investments yield 6%, what is the current value of Cameron Corp. bonds?

Solution: $PMT = \frac{\$1,000 \times 5\%}{2} = \25

$FV = \$1,000$

$N = 9 \times 2 = 18 \text{ periods}$

$I/Y = \frac{6\%}{2} = 3\% \text{ semiannually}$

$PV \text{ (current price)} = \931.23

Section 10.2 LO 1 Moderate

8. McLaren Motors just issued a series of \$1,000.00 bonds with a 10-year maturity and an 8% coupon rate, paid quarterly. If you purchase a McLaren bond at a price of \$920.00, what is your required rate of return?

Solution: $PMT = \frac{\$1,000 \times 8\%}{4} = \20

$FV = \$1,000$

$N = 10 \times 4 = 40$

$PV = -\$920$

$I/Y = 2.3085\%$

$\text{Annualized YTM} = 2.3085\% \times 4 = 9.23\%$

Section 10.2 LO 1 Difficult

9. Three years ago, Petty Partners Inc. issued 15-year, \$1,000 bonds that are currently priced at \$911.37. If the prevailing rate of return on similar investments is 5%, what is the coupon rate on Petty Partners bonds, and what is the annual interest payment?

Solution: $I/Y = 5\%$

$FV = \$1,000$

$N = 15 - 3 = 12$

$PV = -\$911.37$

$PMT = \$40$

The annual coupon payment is \$40, making the coupon rate $\frac{\$1,000}{\$40}$, or 4%.

Section 10.2 LO 1 Difficult

10. A \$1,000 Riker Corp. bond has a 20-year maturity and a 6% coupon rate, with interest paid annually. If similar bonds from Riker Corp. are yielding 4%, what is the current market value of the Riker issue?

Solution: $FV = \$1,000.00$

$N = 20$

$I/Y = 4\%$

$PMT = 6\% \times 1,000 = 60.00$

$PV = \$1,271.81$

Section 10.2 LO 1 Moderate

11. A \$1,000 bond that matures in eight years, has quarterly coupon payments of \$25, and is currently priced at \$962.00 will have a yield to maturity of _____.

Solution: $FV = \$1,000.00$

$PV = -\$962.00$

$N = 8 \times 4 = 32$

$PMT = \$25$

$I/Y = 2.68\% \times 4 = 10.72\%$

Section 10.2 LO 1 Difficult

Chapter 10

Test Bank

Bonds and Bond Valuation

True or False Questions

1. 1. A “junk” bond is one that is speculative grade, with a rating below BBB-.

- A. True
- B. False

Solution: A. True

Section 10.4 LO 4 Easy

Multiple Choice Questions

2. A bond might be thought of as form of what?

- A. Derivative
- B. Stock
- C. Loan
- D. Security

Solution: C. As a form of borrowed money, in this case from the market, a bond is similar to a loan.

Section 10.1 LO 1 Easy

3. Why are bonds commonly referred to as fixed-income securities?

- A. The return on a bond is guaranteed.
- B. Most bonds pay a fixed rate of interest.
- C. The maturity date of a bond is specified.
- D. A bond's yield to maturity cannot change.

Solution: B. *Fixed income* refers to the common feature of all bonds: an unchanging coupon interest rate.

Section 10.1 LO 1 Moderate

4. The interest rate paid by a bond issuer is referred to as the bond's what?

- A. Coupon rate
- B. Yield to maturity
- C. Collateral
- D. Indenture

Solution: A. The coupon, or interest, rate is the percentage of the bond's face value paid out annually to the bond owner.

Section 10.1 LO 1 Moderate

5. A bond's yield to maturity is the same as what?
- A. The yield available in the market for comparable bonds
 - B. The factor used to discount the bond's coupon payments and principal at maturity
 - C. The return earned by the bondholder if the bond is held to maturity
 - D. All of the above

Solution: D. A bond's YTM is a market rate that also serves the mathematical functions of discounting the bond's cash flows and determining the return on bonds held to maturity.

Section 10.1 LO 1 Difficult

6. A municipal bond is issued by which of the following?
- A. A state or local government
 - B. The federal government
 - C. A corporation
 - D. An individual

Solution: A. Municipal bonds, or munis, are issues by state or local governments or their agencies.

Section 10.1 LO 1 Easy

7. If a bond has a call feature, what does that mean about the bond?
- A. It is secured by collateral.
 - B. It has a fixed rate.
 - C. The bondholder can choose to sell it back to the issuer.
 - D. The issuer can choose to buy it back from the bondholder.

Solution: D. A call feature is a benefit to the bond issuer, who can decide, based on market conditions, if they would like to repurchase the bond.

Section 10.1 LO 1 Difficult

8. Corporate bonds are typically issued with a fixed coupon interest rate. What kind of bond does **not** pay a regular interest payment?
- A. Floating-rate bond
 - B. Zero-coupon bond
 - C. Derivative bond
 - D. Junk bond

Solution: B. Zero-coupon bonds do not pay a regular coupon payment. Instead, they are issued at a discount to face value and are paid at maturity at the full face value.

Section 10.1 LO 1 Moderate

9. What is the relationship between a bond's price and the prevailing interest rates for comparable bonds?
- A. Direct
 - B. Inverse

- C. Neutral
- D. Unrelated

Solution: B. A bond's price moves in the opposite direction of the market yields for comparable bonds. Thus, a bond's market price increases when yields decrease.

Section 10.1 LO 1 Moderate

10. What is calculated by multiplying the face amount of a bond by its coupon interest rate?

- A. Annual coupon payment
- B. Yield to maturity
- C. Current yield
- D. Discount rate

Solution: A. Multiplying the face amount of a bond by its coupon interest rate gives you the annual coupon payment.

Section 10.2 LO 2 Moderate

11. The convention for quoting bond prices is different from price quotes for other instruments, such as stocks. How is a bond's price quoted?

- A. As a function of its coupon rate
- B. As a function of its yield to maturity
- C. As a percentage of its face value
- D. In relation to comparable bonds

Solution: C. A bond's price is quoted as a percentage of its face value.

Section 10.2 LO 2 Moderate.

12. For an individual borrower, such as the US government, what happens to bond yields as the bonds mature and lengthen?

- A. They increase.
- B. They decrease.
- C. They remain the same.
- D. The relationship is indeterminable.

Solution: A. As depicted in an issuer's yield curve, the issuer's yield increases as bond maturities lengthen, primarily because longer-dated bonds are riskier than shorter ones.

Section 10.2 LO 2 Difficult

13. When making an investment choice between a stock and a bond, what is an advantage of the bond?

- A. Easier price discovery
- B. Better return
- C. Easier transaction execution
- D. Lower risk

Solution: D. Because bonds are a debt instrument, they have lower risk than stocks, as bond issuers are contractually obligated to pay interest as prescribed and repay the principal at maturity. Stocks have no such features.

Section 10.2 LO 2 Moderate

14. What is a particularly attractive feature of a municipal bond?

- A. It is default free.
- B. Its interest payments are tax advantaged.
- C. It is convertible into stock.
- D. Its market price does not change.

Solution: B. Interest received from a municipal bond is not taxable for federal purposes, and sometimes for state and local purposes too.

Section 10.2 LO 2 Moderate

15. When a bond's coupon interest rate is greater than its yield to maturity, how will the bond sell?

- A. At a premium
- B. At a discount
- C. At par value
- D. Cannot be determined

Solution: A. A coupon higher than market yield indicates that the bondholder can sell the better-performing bond at a premium to par.

Section 10.2 LO 2 Difficult

16. What is a yield curve?

- A. A graph of the yields of various bonds over many maturities
- B. A graph of many issuers' debt maturities
- C. A comparison of a debt issuer's risk-versus-return trade-off
- D. A graph of bond prices at varying yields

Solution: A. The yield curve is a picture of the yields of various comparable bonds over different maturity periods.

Section 10.3 LO 3 Difficult

17. What is one important function of the yield curve?

- A. It allows comparison of an issuer's debt to the overall market.
- B. It predicts interest rate levels in the future.
- C. It compares the riskiness of bonds.
- D. It helps determine a bond's yield to maturity.

Solution: B. By showing a range of debt maturities, from very short to long term, the yield curve gives useful information about interest rates in the future.

Section 10.3 LO 3 Moderate

18. An inverted yield curve is generally interpreted to mean what?

- A. Short-term rates are higher than long-term rates, which may indicate an oncoming recession.
- B. Long-term rates are higher than short-term rates, as is the normal market condition.
- C. Interest rates will be mostly unchanged into the future.
- D. Yield curves do not change, so interest rate inferences are not possible.

Solution: A. An inverted yield curve indicates that long-term rates have fallen below short-term rates, generally indicating an oncoming economic decline that may require a response from monetary authorities to ease rates.

Section 10.3 10.4 LO 4 Difficult

19. When investors believe that a bond issuer is unlikely to meet its payment commitments, this is called _____ risk.

- A. liquidity
- B. reinvestment
- C. default
- D. duration

Solution: C. Default is nonpayment of a debt obligation, so default risk is the risk of a bond issuer not paying its debt.

Section 10.4 LO 4 Easy

20. What is the primary activity of bond-rating agencies?

- A. Drafting bond indentures
- B. Acting as paying agents for bond coupon payments
- C. Providing tax advice on municipal bonds
- D. Evaluating the financial strength of a bond issuance

Solution: D. Rating agencies evaluate the issuer and issuance of a bond and then assign a rating according to the agency's formulas.

Section 10.4 LO 4 Moderate

21. What is a key difference between a rating of BBB- and a rating of BB+?

- A. They differ by half a grade rating level.
- B. BBB- bonds are classified as investment grade, while BB+ bonds are speculative grade.
- C. Bonds with a BB+ grade may not be purchased by many institutional investors.
- D. No significant difference exists.

Solution: C. When an issuance is rated BB+ or below, a significant number of institutional investors are precluded by their investment policy from buying such instruments.

Section 10.4 LO 4 Moderate

22. The strategy known as *bond laddering* consists of what?

- A. Buying multiple issuers' bonds with the same maturity date
- B. Buying bonds with many different maturity dates
- C. Buying bonds that will be sold at specified dates
- D. Converting bonds into equity when they are called

Solution: B. Buying bonds with multiple different maturity dates spreads risk and cash flows among a portfolio of bonds.

Section 10.4 LO 4 Moderate

23. A bond issuer is more likely to call the bond if which of the following happens?

- A. Prevailing interest rates fall.
- B. Prevailing interest rates rise.
- C. Tax rates rise.
- D. The issuer's credit rating declines.

Solution: A. If interest rates fall, the issuer may decide to call the bonds and issue new bonds at a lower all-in cost.

Section 10.4 LO 4 Difficult

24. For an investor, why are bonds attractive?

- A. They pay regular, predictable interest.
- B. Their returns are normally higher than stock.
- C. They can be purchased almost anywhere.
- D. They are not correlated with stock prices.

Solution: A. Bonds are particularly attractive to investors looking to receive the regular income they provide.

Section 10.4 LO 4 Moderate

25. When calculating the price of a bond using a calculator or Excel, the amount of each interest payment would be entered as what?

- A. PV
- B. FV
- C. PMT
- D. YTM

Solution: C. PMT is the payment made to the investor, representing a portion of the bond's value/price.

Section 10.5 LO 5 Moderate